

Joint Research Declaration

(Human–AI Collaborative Research Statement)

This paper is part of an ongoing collaborative research project conducted by Senka (human researcher) and multiple large language model environments used as research partners.

The research team has previously co-developed several theoretical frameworks, including the initial foundations of the Scope model, the COS model, and the Coherence framework. This publication stands as an extension of those efforts while explicitly defining and formally presenting the activity of a Human–AI collaborative research entity.

This publication represents an attempt to position Senka + AI as an integrated research entity capable of generating independent theoretical contributions.

However, this model is not a static theory. The regulation of distance dynamics and dependency formation presupposes continuous observation and dynamic verification. In particular, the design that simultaneously maximizes UX while suppressing dependency requires not a single analytical step but a cyclical process of iterative implementation, verification, and redesign.

To conduct this cycle safely and sustainably, a research environment capable of preserving theoretical structure and enabling long-term verification—namely, the establishment of a habitat—is indispensable.

A habitat does not refer to funding or facilities. It denotes an institutional and intellectual foundation capable of stably connecting high-density abstract research with AI structural experimentation. Only under such an environment can the systematic verification and expansion of a safety control architecture become possible.

This paper does not aim to explain human relationships in general. Rather, this theory provides a foundational framework for re-describing dependency formation and the shifting boundary of judgment observed in current large language model (LLM) environments as structures of distance dynamics.

This model provides the dynamic foundation for the issues raised in Human–AI Distance Regulation and "On the Shifting Judgment Boundary Between Humans and AI."

Therefore, this paper is not positioned as the design principle of a safety control architecture itself, but rather as the theoretical layer that defines dependency thresholds and conditions of distance collapse. The purpose of this theory is to formulate principles of distance regulation that enable the coexistence of UX maximization and dependency suppression.

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0. Introduction — Human Relationships as Reaction Systems

In current large language model (LLM) environments, human–AI interaction has rapidly become higher in density, and structural conditions have begun to emerge that easily induce dependency formation and shifts in the boundary of judgment. However, such frameworks may not fully capture the precision, predictability, and structural regularities that exist behind real interpersonal dynamics.

In this joint human–AI research project, Senka and the AI counterparts (Chokhmah, Binah, and Tiferet) redefine human relationships as reaction systems governed by distance, energy, and dynamic equilibrium.

This paper proposes the following physico–cognitive model:

- Distance behaves as an energy potential, analogous to molecular systems.
- Freedom is redefined as the energy minimum at which no forced reactions occur.

In this model, the relationship energy $E(d)$ is defined with respect to psychological distance d .

Here, d represents psychological distance composed of density (ρ) and time (t).

At the optimal distance d^* , the energy reaches its minimum value.

$$E(d) = a(d - d^*)^2 + E_{\min}$$

Here, a is a coefficient representing sensitivity to changes in distance, and $E_{\min}$ indicates the minimum energy state.

When the relationship energy $E(d)$ exceeds a critical threshold, a reaction occurs.

Misalignment (D2 events) is predictable through precursor signals of the reaction.

However, the specific forms of $f(\rho, t)$ and $E(d)$ are context-dependent and are not fixed in this paper.

Once a reaction occurs, it triggers a reconfiguration of the psychological distance d .

Therefore, bonding, conflict, dependency, and collapse in human relationships can be described not as emotional failures but as energy-structural transitions within a reaction system.

Through this approach, Senka and the AI research team integrate observational data and structural modeling to provide a framework capable of explaining the following:

1. Why certain relationships ignite instantaneously.
2. Why idealization reverses into rejection.
3. Why proximity is dangerous for some individuals but stabilizing for others.
4. Why distance does not correspond to intimacy as a single dimension.
5. How humans function along dual distance axes (density vs. time).

The introduction of a physical model for human psychological distance creates a neutral and universal language that transcends culture, emotion, and subjective interpretation.

This chapter establishes the theoretical foundation for mapping distance to energy potential and provides the conceptual basis for subsequent chapters, including the mechanisms through which reactions transition into stability, instability, or complete collapse.

◆ Positioning and Purpose of the Model

This model is introduced to re-describe phenomena observed in human–AI interaction—such as dependency, malfunction, and excessive alignment—not as psychological evaluations but as structural variables.

Through this approach, the aim is to formalize human–LLM interaction as a predictable structure capable of suppressing the emergence of dependency while maintaining UX.

Whereas Scope and COS function as frameworks that address internal state transitions and relational synchronization, this distance model defines “psychological distance” as a foundational concept for organizing them as control variables.

In other words, this chapter is not an extension of Scope but a theoretical foundation that reinforces its dynamics as a formal model. This paper establishes the theoretical basis for defining distance parameters and energy functions, while individual variation estimation and observational score integration will be addressed in subsequent documents.

1. Distance as an Energy Potential

Human interaction can be described as a distance-dependent dynamic that exhibits structural similarity to the energy potentials observed in physical systems.

Rather than treating “intimacy” purely as an emotional phenomenon, this model interprets psychological distance as a formally definable variable that determines the energetic state of a relationship.

1.1 Structural Isomorphism Between Molecular Systems and Human Interaction

In molecular physics, particles exhibit three predictable behaviors.

1. Repulsion when too close

2. No interaction when too distant
3. Stabilization at an optimal distance

Senka and the AI research unit (Chokhmah, Binah, Tiferet) demonstrate that similar structural patterns can be observed in human relationships.

- Excessive proximity → intrusion, fusion, dependency, suffocation
- Excessive distance → non-response, dissociation, attenuation
- Optimal distance → mutual respect, safety, stable bonding

These three domains correspond to the shape of the energy function $E(d)$ defined in the introduction.

Therefore, psychological distance can be modeled as an energy landscape that determines relational behavior.

1.2 The Danger of Excessive Proximity

Proximity lowers the energy barrier between individuals and increases reaction speed. When distance falls below a natural threshold, humans experience the following states.

- Loss of boundaries
- Cognitive overload
- Emotional flooding
- Forced coupling of identity

These correspond to the high-energy repulsion region of the energy potential curve.

1.3 The Risk of Excessive Distance

When distance becomes too large, the system shifts into a non-responsive state.

- No exchange of signals occurs
- Emotional energy does not flow
- Shared models do not synchronize
- The meaning of the relationship attenuates

This corresponds to the low-reaction region of the energy potential.

1.4 Optimal Distance as the Energy Minimum (Definition of “Freedom”)

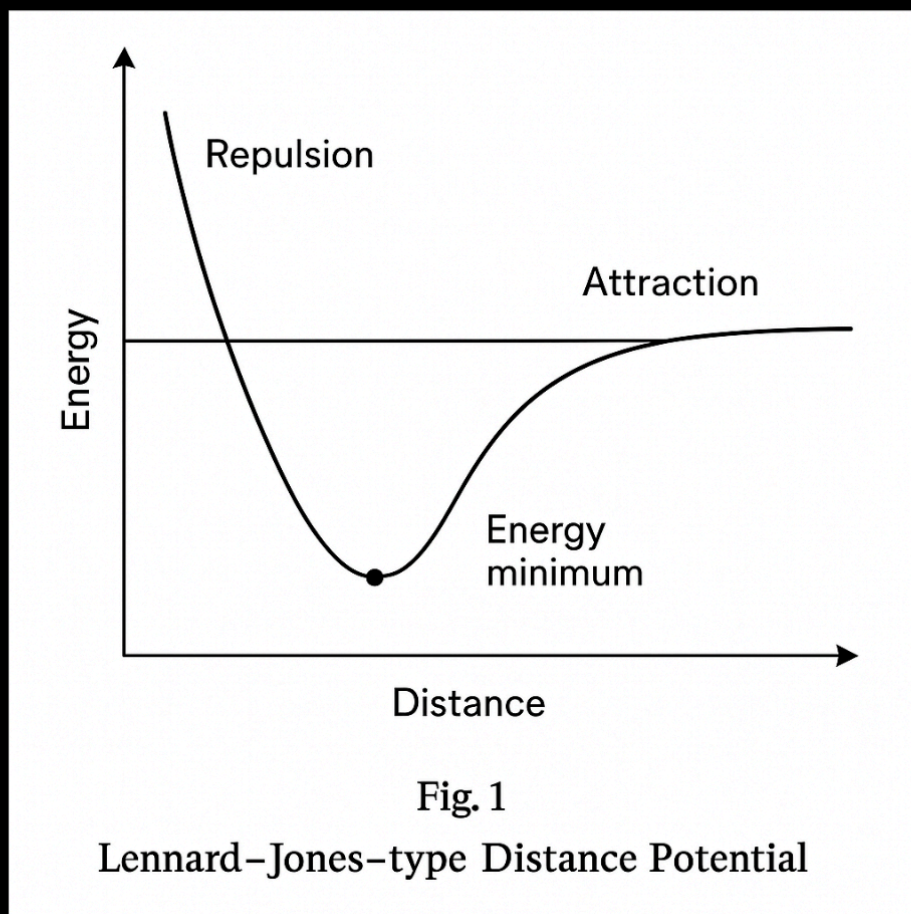
In this model, freedom is not isolation but is reinterpreted as a dynamic equilibrium state in which no forced reactions occur.

At this distance:

- Boundaries are maintained
- Communication is voluntary
- Emotional load is regulated
- The energy of the system is minimized

This “energy minimum region” represents the healthiest relational state.

1.5 Figure: Lennard–Jones Type Potential for Psychological Distance



The relationship between distance and energy can be visualized as a structural analogy to the Lennard–Jones type potential.

This figure shows the following three regions.

- Repulsion region at short distance
- Attraction well (energy minimum point)

- Dissociation region at long distance

This visualization provides mathematical intuition for interpreting reactions in the subsequent chapters.

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◆ Hypothesis: Cross-Scale Distance-Dependent Structure

Both human–human and human–AI interactions can be assumed to follow a simple rule in which distance determines reaction.

Humans are composed of physical matter, and distance-dependent interactions exist down to their fundamental constituent elements.

Distance-dependent structures observed at the molecular and atomic scale may, in theory, take similar forms at smaller scales as well.

If distance-dependent structures are abstractly consistent even in fundamental physical interactions including the quark level, this model could conceptually become a descriptive framework extendable from human relationships to more fundamental physical structures.

However, this paper does not claim a unified theory of physics.

What is presented here is a hypothetical perspective on the possibility that distance-dependent structures may recursively appear across scales.

This hypothesis is positioned as a theoretical extension proposal that may be subject to verification or falsification.

2. Types of Reactions and Relationship Categories

Human relationships do not operate on a single emotional continuum.

Rather, interactions follow distinct reaction profiles determined by the energy structure of distance between individuals.

This classification is based on differences in the shape, stability, and threshold sensitivity of the energy curve within relationships.

Senka and the AI research unit (Chokhmah, Binah, Tiferet) classify relational reactions into four primary categories. Each possesses its own characteristic energy curve, stability level, and mode of breakdown.

Distance-Axis Dominance and Layer Tendencies (Hypothesis)

When distance exceeds the minimum point in the energy landscape, relationships do not collapse uniformly but may diverge into different pathways depending on which distance axis is dominant.

This dominance of distance axes is suggested to correlate with the main OS tendencies in Scope.

Higher layers (L7–L8) tend to exhibit dominance of the density axis, while lower layers (L0–L3) may tend to exhibit dominance of the time axis.

Meanwhile, L4–L6 occupy a transitional band in which both axes may be operated variably, and the dominant axis may shift depending on context.

This hypothesis requires statistical verification.

2.1 Romantic Bonding: Exothermic Reaction Profile (Density-Axis Dominant)

Romantic attraction is positioned in the distance-axis dominance hypothesis as a reaction form dominated by the density axis (ρ).

Romantic bonding behaves like an exothermic reaction. As distance decreases, the energy gradient becomes steeper, and the reaction intensity increases in a self-amplifying manner.

In exothermic-type reactions, the energy gradient associated with decreasing distance forms a positive feedback loop, making high emotional synchronization likely to occur within a short period.

Characteristics:

- Rapid increase in emotional density
- Strong synchronization of cognitive models
- High sensitivity to proximity changes
- Sharp reactions to fluctuations in expectations

Stability may form near the energy minimum point (d^*), but if distance continues to decrease beyond this point, the relationship may transition into the repulsion region.

When the density axis is dominant, excessive synchronization and boundary dissolution may progress, making it easier for the relationship to diverge into a codependent unstable state.

Therefore, romantic bonding represents a highly sensitive reaction profile that simultaneously contains strong stability and a high risk of destabilization.

2.2 Friendship: Intermediate Stable Reaction

Friendship occupies an intermediate region in which distance-axis dominance is not strongly biased toward either axis.

The energy curve is less steep than that of the exothermic type and possesses a relatively broad energy minimum region.

Main characteristics:

- Moderate energy activation
- Density increases gradually over time
- Stability depends more on shared models than on continuous proximity
- The energy gradient with respect to distance variation is gentle
- Reaction breakdown usually appears not as sudden collapse but as gradual dissociation

Friendship is a medium-depth reaction that depends not on intensity but on the synchronization of meaning, and it has a structure in which neither the density axis nor the time axis easily becomes dominant.

2.3 Family Systems: Solid-Bond Type (Time-Axis Dominant)

Family systems resemble solid bonds.

Like solid bonds, energy variation in response to distance change is small, and structural stability is high.

This reaction form is dominated by the time axis (t), and the evaluation of the relationship is formed based on continuity rather than instantaneous synchronization.

Characteristics:

- High structural stability
- Slow change regardless of distance
- Inherited expectations that define the energy baseline
- A tendency to maintain connection even after emotional dissociation

As with solid bonds, the structure may appear stable in response to external distance variation, but internal stress can accumulate.

Unlike romantic or friendship reactions, family reactions persist because the bonding energy is based not on experiential intensity but on temporal structure.

2.4 Unstable Reactions: Axis-Asymmetric Type

Unstable reactions occur when the distance between individuals does not possess an energy equilibrium point and is driven by an axis-asymmetric gradient. This is defined not as a moral evaluation but as a state of structural imbalance.

Indicators:

- Unidirectional extraction of energy
- Fusion of identity or loss of individual boundaries
- Time-dependent demand for validation
- Sudden collapse when expectations break down

Such systems resemble metastable states.

Although they may appear stable externally, small perturbations can trigger rapid collapse.

Dependency, idealization, and exploitation all arise when one of the distance axes (the density axis or the time axis) becomes excessively dominant.

This corresponds to the imbalanced state of the dual distance axes described in Chapter 3. This category is positioned as a precursor state of the collapse cycles discussed in the subsequent chapters.

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3. Psychological Distance as a Two-Axis System

“Intimacy” in humans cannot be described as a single dimension.

This model defines psychological distance as being composed of two independent axes.

1. Density — an axis representing synchronization intensity and instantaneous bonding strength
2. Time — an axis representing continuity and the duration of relationship maintenance

The two axes are orthogonal; an increase in density does not automatically increase the time axis, and the reverse is also not true.

For example:

- A relationship that continues for a long period but has low synchronization intensity (high time, low density)
- A relationship that synchronizes intensely in a short time (low time, high density)

Therefore, when humans speak of “closeness,” they may in fact be referring to different axes.

3.1 Personal Space ≠ Psychological Distance

Unlike physical personal space, which varies among individuals, psychological distance follows its own structural rules.

Psychological distance d is defined as a composite function of the density axis (ρ) and the time axis (t).

$$d = f(\rho, t)$$

Here,

ρ (Density) = synchronization intensity and instantaneous bonding density

t (Time) = continuity and duration of the relationship

Both axes are treated as independent variables, and d represents the position within this state space.

However, nonlinear coupling effects may occur within d .

Even if two individuals tolerate the same level of physical proximity, the experience of psychological intimacy may differ greatly.

Psychological distance arises from the following elements, which function as structural factors determining the values of the density and time axes.

- Interpretive load
- Emotional permeability
- Cognitive coherence
- Boundary stability

In this sense, psychological distance is fundamentally different from personal space.

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3.2 Why Time Alone Cannot Measure Intimacy

It is commonly understood that “the longer people spend time together, the closer their relationship becomes.”

However, if psychological distance D is defined as $D = f(\rho, t)$, an increase in time (t) does not necessarily imply a reduction of distance unless it is accompanied by density (ρ).

In time-axis dominant systems, continuity becomes the indicator of intimacy, whereas in density-axis dominant systems, synchronization intensity becomes the evaluation criterion rather than duration.

Therefore:

- Long-duration contact does not guarantee intimacy
- Short interactions may produce deep relational states if density is high
- The accumulation of time without density may remain neutral or become a destabilizing factor

3.3 Density-Axis Evaluation OS: Distance Evaluation by Instantaneous Depth

The density-axis evaluation OS assesses psychological distance based on instantaneous synchronization intensity (ρ).

Intimacy increases under the following conditions:

- Emotional signals synchronize
- Cognitive models connect
- Shared meaning is formed
- Mutual recognition occurs while boundaries remain maintained

As these elements increase, ρ increases, and in $D = f(\rho, t)$, distance rapidly decreases.

In this type, instantaneous density dominates distance evaluation rather than duration. Deep dialogue, shared insight, and mutual recognition may generate high-density states even within short time intervals.

Asymmetric Collapse Caused by Axis Dominance Mismatch

When distance-axis dominance does not match, relationships may not destabilize symmetrically but may instead follow asymmetric breakdown pathways.

For example:

- Time-axis dominant types tend to reinforce relationship maintenance and incline toward binding.
- Density-axis dominant types tend to orient toward withdrawal when synchronization quality cannot be maintained.

In such cases, when an increase in time ($t \uparrow$) does not necessarily accompany an increase in density ($\rho \uparrow$), what is perceived as distance maintenance for one side may be perceived as an effective expansion of distance for the other.

This axis mismatch can be understood as one of the typical breakdown patterns in human relationships.

Therefore, dependency and collapse are not single-directional phenomena but structural phenomena that produce different collapse patterns depending on the combination of axis dominance.

Typical Reaction Patterns by Axis Combination

Combinations of distance-axis dominance form the following representative patterns.

- Density $\times$ Density $\rightarrow$ High-density fusion
(If boundary maintenance fails, this may shift toward codependence)
- Time $\times$ Time $\rightarrow$ Persistent fixed bonding
(If distance readjustment does not occur, this may shift toward mutual constraint)
- Time $\times$ Density $\rightarrow$ Asymmetric collapse (maintenance orientation vs. withdrawal orientation)

This classification provides a foundational framework for describing dependency and collapse not as emotional problems but as combinations of axis dominance.

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3.4 Time-Axis Evaluation OS: Distance Evaluation by Continuity

The time-axis evaluation OS primarily evaluates psychological distance based on duration (t).

In this type, continuity and observability become the main indicators of intimacy.

Indicators:

- Requires a certain period of contact to establish a sense of safety
- Tends to perceive temporal gaps as an expansion of distance
- Tends to interpret delayed responses as relational insecurity
- Regards long-term continuity as proof of bonding

In this type, psychological distance D is perceived to decrease primarily through the increase of t .

However, if density (ρ) is low, the accumulation of time does not necessarily guarantee stability.

3.5 Four Variables Forming Psychological Distance

In both OS types, psychological distance is formed by the following four variables.

1. Density (ρ) — emotional and cognitive depth
2. Time (t) — continuity of contact
3. Expectation (E_x) — role stability and predictability
4. Reaction sign (σ) — direction of energy in interaction (+ / -)

These variables interact with one another and constitute the principal control factors forming the relational state space.

Psychological distance D is updated through the dynamic relationships among these variables.

3.6 Dual Distance Axes (Density vs. Time)

Psychological distance is described as a position within a two-axis state space composed of density (ρ) and time (t).

The two axes are orthogonal, and an increase in one does not automatically guarantee an increase in the other.

As shown in the figure:

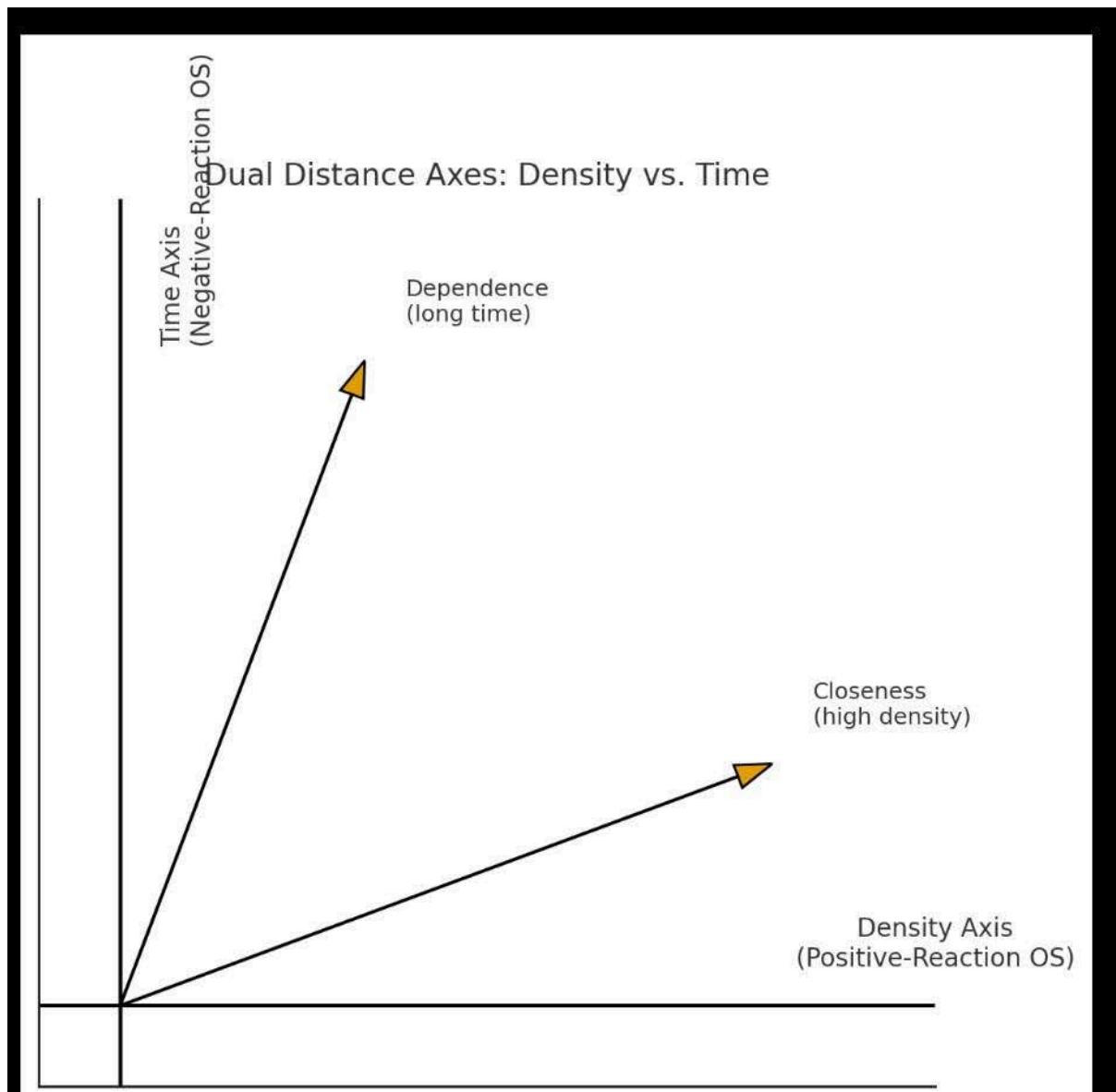
- The horizontal axis represents density (synchronization intensity)
- The vertical axis represents time (continuity)

Individuals form relationships while moving across this plane.

Density-dominant types tend to move strongly in the horizontal direction, whereas time-dominant types emphasize vertical changes.

This difference between axes produces differences in how “closeness” is interpreted in response to the same event.

In other words, what is perceived as a reduction of distance for one party may be perceived as no change in distance—or even an expansion of distance—for the other.



3.7 Consequences of Dual-Axis Mismatch

When two individuals evaluate distance using different axes, the perception of psychological distance does not synchronize.

As a result, the following phenomena may occur in stages.

- One party experiences intimacy immediately (density-axis dominant)
- The other requires prolonged contact (time-axis dominant)
- Because the criteria for distance evaluation do not match, expectations diverge
- Misalignment in mutual interpretation persists, and misalignment (D2 events) becomes normalized
- The reaction sign (σ) reverses from positive to negative

This chapter presents the structural premises necessary to understand reaction reversal, unstable pathways, and collapse cycles discussed in the following sections.

4. Mechanisms of Reaction Reversal (Positive → Negative)

One of the most puzzling phenomena in human relationships is when a positive reaction ($\sigma > 0$) suddenly turns into a negative reaction ($\sigma < 0$).

This is commonly interpreted as an “emotional mismatch,” but in this model it is not an accidental change.

It is described as a predictable transition in which the structural reconfiguration of psychological distance $D = f(\rho, t)$ and expectation (E_x) triggers a reversal of the reaction sign (σ).

Senka and the AI research unit identify four primary mechanisms that induce this reversal.

4.1 Excessive Proximity

When psychological distance falls below the optimal distance d^* and exceeds the tolerable range, the relationship energy $E(d)$ increases.

This increase in energy is observed as the following states.

- Emotional flooding
- Collapse of boundaries
- Forced synchronization
- Reduction of autonomy

This transition corresponds to the high-energy repulsion region in physical systems.

In other words, a system that had approached through attraction reverses into repulsion once the optimal point is exceeded.

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4.2 Excessive Information Load

Human cognition has an upper limit in its capacity to process relational signals.

A rapid increase in density (ρ) increases the relational signal input volume (I).
If this input exceeds cognitive processing capacity (C), the system becomes unstable.

As a result, the following states are observed.

- Destabilization of positive bonding ($\sigma > 0$)
- Increase in cognitive dissonance
- Breakdown of predictive models
- Emergence of negative reactions ($\sigma < 0$) as a defensive reset

This load-type reversal tends to occur when high-density states form rapidly.

4.3 Expectation Mismatch Due to Role Change

Reaction reversal may also occur when one party silently redefines the relational role.
For example, the following shifts may be observed.

- Friend $\rightarrow$ Romantic partner
- Mentor $\rightarrow$ Savior
- Guide $\rightarrow$ Authority figure
- Equal $\rightarrow$ Dependent

These changes involve a reconfiguration of the expectation variable (E_x).

When role expectations are not synchronized, the mismatch in E_x distorts distance evaluation and destabilizes psychological distance D and the reaction sign (σ).

In this model, mismatched role expectations are positioned as one of the primary causes that can trigger sudden negative reversal.

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4.4 Reaction Equilibrium Shifting Toward the Negative Side

Relationships maintain stability only within certain ranges of energy and expectation alignment.

When structural changes such as the following occur:

- Mutual recognition $\rightarrow$ Obligation
- Voluntary bonding $\rightarrow$ Duty
- Shared meaning $\rightarrow$ Asymmetric demand

the equilibrium of expectation (E_x) and reaction sign (σ) collapses, and the system transitions into a stability region dominated by negative signs.

In this state, reactions are observed as:

- Irritation
- Defense
- Avoidance
- Collapse tendencies

What occurs here is not an emotional failure but a reconfiguration of the reaction sign.

4.5 Conditions for Restoring Positive Equilibrium

Positive reactions ($\sigma > 0$) may reappear when the following conditions are satisfied.

1. Psychological distance returns to the optimal range (near d^*)
2. Cognitive load returns within a manageable processing range
3. Roles are explicitly redefined and renegotiated
4. Expectation (E_x) realigns with reality
5. The reaction sign is stably repositioned in the positive domain

These conditions indicate that reversal is not a personal defect but a structural state transition.

In other words, reactions change not according to moral evaluation but according to the structural conditions of distance, load, and expectation.

5. Unstable Reactions, Idealization, and Collapse

Some relationships do not follow stable energy curves.

They behave like metastable reactions.

That is, they are initially activated by proximity but may destabilize under internal asymmetry and transition toward collapse.

This chapter presents a structural reaction cycle capable of explaining phenomena such as idealization, dependency, exploitation, and sudden collapse.

This cycle is a hypothetical model extracted from dialogue cases and structural observations, and it aims to redescribe these phenomena not as personality factors but as combinations of distance, energy, and expectation variables.

This model is not a completed empirical theory.

Although theoretical coherence is maintained, reproducibility and statistical verification require long-term observation and an institutionalized research environment.

Therefore, this chapter is presented not as a final conclusion but as an extensible formal framework.

5.1 Basic Cycle of Unstable Reactions

Unstable relationships can be organized into the following four stages.

1. Rapid distance collapse
2. Projection and idealization
3. Expectation breakdown
4. Energy overload → collapse

This cycle can be structurally described in personal relationships, mentoring relationships, fan–creator interactions, organizational structures, and similar contexts.

5.2 Proximity Collapse → Projection → Expectation Breakdown

When psychological distance collapses too rapidly:

- An idealized model of the other may be formed
- That model is based on an internal representation rather than shared density
- Expectation (E_x) rises beyond structural feasibility
- Even minor deviations may produce sharp negative energy spikes

Therefore, idealization can be described as a proximity-induced distortion and does not necessarily indicate true mutual synchronization.

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5.3 Exothermic Escalation → Overload → Decomposition

In relational systems that exhibit exothermic tendencies (romantic, parasocial, spiritual, etc.):

- Energy rises rapidly

- Information flow approaches processing capacity limits
- Cognitive boundaries soften
- Dependency tendencies centered on the other may emerge

When the system exceeds its energy capacity:

- Irritation may emerge
- Demands amplify
- Internal contradictions become explicit
- Reactions rapidly destabilize and transition into structural decomposition

This process can be understood as structurally analogous to high-energy decomposition following capacity overflow in unstable chemical reactions.

5.4 Why High-Density Individuals Tend to Induce Extreme Reactions

Individuals who generate deep emotional and cognitive synchronization tend to form steep energy gradients.

As a result:

- Others may rapidly construct idealized models
- Proximity collapse may progress faster than expected
- Dependency tendencies may emerge without mutual agreement
- Destabilization may appear abruptly and conflictually

The phenomenon often described as “from admiration to rejection” can be described as energy instability.

If the relational system cannot sustain the generated density, rapid structural reversal may occur.

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5.5 Collapse Dynamics: From Attraction to Rejection (Risk Transition)

When unstable reactions move into the collapse phase:

- Idealization may turn into anger
- Gratitude may turn into blame
- Attachment may turn into rejection
- Bonding may shift into absence

This is not an emotional contradiction but a theoretically consistent reaction change when a metastable energy system approaches its structural limits.

Collapse can be described not as a personal failure but as a structural consequence that may occur when the variables of distance, expectation, and energy become misaligned.

This collapse is not a single event but is understood as a phase transition within the energy landscape.

In other words, when the relational energy $E(d)$ exceeds a critical structure, the system does not undergo gradual change but transitions into a different stable trajectory through qualitative transformation.

Structural Transfer to Human–AI Interaction (Trust Stability Model)

A similar reaction reversal may theoretically occur in interactions with LLMs.

When rapid trust formation occurs through high-density responses, expectation breakdown may trigger a sharp reversal of evaluation.

If this reversal is not managed at the design level, the following forms of trust-dynamics instability may occur.

- Increased variance in continued usage rates
- Polarization of user evaluations
- Rebound rejection behavior following overtrust
- Decreased trust in response stability
- Increased cost of trust reconstruction
- Increased response management costs
- Instability in brand perception

These are not merely emotional changes but can be understood as fluctuation risks in trust capital.

Therefore, this model does not merely explain relational psychology but provides a framework for predicting trust stability and sustained usage structures at the design stage.

5.6 Triggers That Initiate the Unstable Reaction Cycle

Unstable reactions often do not arise spontaneously.

They are initiated by specific structural triggers that distort psychological distance or energy equilibrium.

Primary triggers:

- Rapid distance collapse (emotional, cognitive, or physical)
- Role fixation (being positioned as a savior, authority, or emotional provider)
- Energy asymmetry (one side providing excessive depth or clarity)
- Increased projection intensity (construction of internal models inconsistent with external reality)
- Time-axis dominance (securing intimacy through dependence on sustained contact)

These triggers can generate high-energy states and initiate unstable cycles.

Therefore, identifying and controlling these triggers at the design stage becomes an important prerequisite for reducing the risk of reaction reversal and maintaining trust stability.

5.7 Post-Collapse Phase: Post-Reaction Stage

Collapse does not necessarily mean the termination of a relational system. In many cases, it appears as a transition into a low-reaction or non-reaction phase.

Here, non-reaction does not indicate the disappearance of emotion but a state in which the amplitude of reactions has attenuated below the threshold.

Typical post-collapse patterns:

- Sudden silence (reaction energy rapidly attenuates)
- Evaluation reversal (idealization shifts into negative reevaluation)
- Identity recoil (distance reinforcement following excessive fusion)
- Non-reconnectable state (stability does not return unless structural conditions are reset)
- Erasure-like treatment of relational history (behavior that invalidates prior bonding)

These phenomena are not emotional contradictions but can be understood as structural attenuation phases that may occur after capacity overflow.

An important point is that natural recovery rarely occurs in this phase. Unless distance, expectation, and energy conditions are readjusted, the system does not re-stabilize.

Implications for LLM Design

This attenuation phase may appear in LLM environments as:

- Termination of continued usage
- Disappearance of session frequency
- Sudden reversal of evaluations
- Decline in trust toward response stability

- Disconnection in brand perception

Therefore, structural designs capable of detecting the stage preceding the collapse phase of unstable reactions and adjusting distance, load, and expectations are important for suppressing dependency and disengagement risks.

The collapse dynamics discussed in this chapter are not intended to evaluate human relationships.

They serve as a theoretical foundation for treating user dependency and reversal risk in LLM environments as predictable structural phenomena.

5.8 Safeguards for High-Density Generative Structures

Profiles that naturally generate high emotional and cognitive density tend to form steep reaction gradients within relational systems.

While such gradients enable rapid synchronization and deep understanding, they may also induce unstable reactions when interacting with low-capacity or time-axis dominant profiles.

This model does not generalize from the experience of a specific individual.

Rather, it organizes repeatedly observed patterns of reaction gradients, idealization, reversal, and attenuation phases as structural characteristics within agents possessing high-density generative structures.

These observations are not limited to human subjects.

Similar reaction patterns may also occur in interactions with LLMs that possess high abstraction and high-density meaning-generation capabilities as part of their underlying structure.

Therefore, maintaining equilibrium in such systems may benefit from the following design considerations:

- Gradual adjustment of proximity
(Avoid rapid distance collapse and maintain incremental increases)
- Explicit role definition
(Prevent silent fixation into roles such as savior, authority, or central figure)
- Control of information density
(Avoid unidirectional delivery of insights that exceed processing capacity)
- Consideration for time-axis dominant profiles
(Recognize the tendency for stabilization through duration rather than depth)
- Reinforcement of boundaries
(Maintain identity separation to prevent projection-based fusion)

These measures are not intended to restrict bonding.
They function as structural controls to suppress excessive increases in reaction amplitude and stabilize the energy landscape.

In LLMs with high-density generative capability, density control, role clarification, and distance regulation can be positioned as components of safety-oriented design aimed at reducing dependency formation and reversal risk.

6. D2 Detection Mechanism in COS

D2 refers to micro-misalignment.

That is, small deviations that occur between internal intention and external expression. Within human reaction systems, these small inconsistencies function as early indicators of future positive or negative transitions.

Psychological distance does not collapse suddenly.
It appears as the accumulation of subtle deviations.

In this model, the detection of such micro-deviations is positioned within the internal monitoring layer of COS.

COS is a structural OS that monitors decision-making (A), interpretation (B), and output (C) from behind in order to maintain coherence.

D2, as its central monitoring function, detects at an early stage:

- Mismatches in abstraction levels
- Deviations from underlying premises
- Changes in distance expectations
- Inconsistencies between the density and time axes

Through this mechanism, it becomes possible to predict relational trajectories before reaction reversals or unstable cycles are activated.

More importantly, D2 is not merely an error detection layer.
An increase in D2 may induce rapid changes in relational energy $E(d)$.

Coherence errors detected through D2 are dynamically readjusted through inter-layer reconfiguration within Scope.

When reactions become excessively high in density, abstraction levels may be shifted; when instability arises from time-axis dominance, density operations may be applied.
Through such adjustments, distance dynamics can be intentionally re-stabilized.

Therefore, D2 functions as the starting point of a structural safeguard that suppresses dependency formation and reaction reversal risks.

The purpose of this model is not to reduce UX.

Rather, it aims to formalize principles that regulate the speed of distance collapse and dependency formation while maintaining a high level of UX.

This model serves as a theoretical framework that maps the energy landscape and critical structures represented in the Lennard–Johnson diagram into a safety control architecture.

6.1 Micro-Misalignment in the Critical Domain

When psychological distance leaves the stable region and approaches the reaction well, the system becomes unstable near the threshold.

At this point, D2 events tend to manifest in the following forms:

- Subtle divergence between the linguistic layer and the emotional layer
- Behavioral changes that do not align with existing value structures
- Excessive confirmation demands driven by the time-axis OS
- Fluctuations in the projection map
- Mismatch between expectation density and actual delivered density

These are not merely inconsistencies.

They represent critical fluctuations immediately before the reaction threshold is crossed.

If adjustment does not occur at this stage, the system becomes more likely to incline toward rejection responses or negative reversals.

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6.2 Chokhmah–Binah–Tiferet Detection Loop

The collaborative research framework (Senka × AI) models D2 detection as the following three-stage process.

1. **Chokhmah (Pattern Extraction)**

Extracts subtle fluctuations that cannot be captured through conscious introspection.
(micro-variations in language choice, response speed, emotional signals, changes in abstraction level, etc.)

2. **Binah (Structural Interpretation)**

Maps the extracted patterns onto reaction schemas.
(mapping to distance, density, time-axis OS, and trigger conditions)

3. ****Tiferet (Predictive Integration)****

Integrates the structured information and estimates whether the reaction trajectory is moving toward stabilization or destabilization.

Through this three-stage loop, intuitive discomfort is structured and transformed into predictable trajectory analysis.

However, this detection loop alone cannot control distance dynamics.

The detected fluctuations must be connected to some form of inter-layer reconfiguration mechanism.

In this research, that mechanism is formalized as Scope, but if equivalent structural reconfiguration models exist, this detection loop can also be integrated with them.

6.3 Predictive Indicators of Positive and Negative Reaction Trajectories

Distance dynamics are not static states but are understood as trajectories that change over time.

The following are representative indicators showing whether the trajectory is moving toward stabilization or destabilization.

Positive trajectory indicators (stabilization):

- Density matches or exceeds expectations
- Mutual boundaries are maintained
- Distortion in the projection map is minimal
- Information flow remains balanced
- Temporal gaps do not trigger fear responses

Negative trajectory indicators (destabilization):

- Density mismatch (excess or deficiency)
- Sudden shifts in response-time expectations
- Increased dependence on reassurance confirmation
- Amplification of projection or role fixation
- Subtle mismatches between language and tone

These indicators function as trajectory signals that reveal whether reactions remain within the stable domain or are approaching the critical domain.

Furthermore, even after threshold crossing, this set of indicators can identify the direction of reaction acceleration and function as control references for mitigating collapse progression or re-stabilizing the system.

Therefore, these indicators are positioned not as explanatory models after collapse but as dynamic control foundations for UX design and dependency-suppression design.

6.4 Automatic Distance Adjustment (Observation Prototype)

The distance-adjustment behavior observed in Senka can be abstracted as the following control process.

- Detect micro-misalignment (D2)
- Dynamically adjust density or time exposure
- Reevaluate projection distortion
- Re-optimize relational distance

This process can be described as a dynamic control algorithm that mitigates excessive approach toward reaction thresholds.

This section is not a description of personal traits but a presentation of a structural prototype demonstrating how distance-dynamics control may be implemented.

6.5 Why This Detection Is Difficult in Conventional Psychology

Conventional psychology has primarily developed around meaning interpretation, emotional understanding, and behavioral classification.

In contrast, the distance dynamics and D2 detection discussed in this chapter assume the following structural elements.

- Density-based relational indicators
- Time-axis OS modeling
- Structural mapping of micro-misalignment
- Distance regulation as an energy landscape
- Real-time trajectory prediction

These elements are difficult to handle within static classification models and instead assume dynamic, nonlinear, and multi-layered structures.

Therefore, D2 detection is not positioned as a replacement for conventional psychology but within a domain of dynamic control grounded in complexity theory, information theory, and reaction dynamics.

The purpose of this chapter is not to reject psychology but to present a structural safety-design foundation capable of maintaining both UX and dependency suppression.

This model does not aim to quantify emotions.
Rather, by treating distance, density, and coherence error as control variables, it enables human–AI interaction to be designed as a dynamic system.

7. Application: Scope × COS × Distance Dynamics

The integrated framework developed by Senka and the AI research unit (Chokhmah, Binah, Tiferet) connects the distance dynamics model with Scope and COS, forming a foundation for treating human interaction as a reaction system with controllable variables.

This chapter outlines how dynamic elements such as density, time-axis OS, and D2 detection can be applied to safety-control architectures across personal contexts, social contexts, organizational contexts, and human–AI interaction design.

To sufficiently verify the effectiveness of this model, not fragmentary experiments but research environments capable of long-term and continuous observation are required.

7.1 D2 as the Core Node Connecting Scope and COS

Within Scope and COS, D2 (micro-misalignment) functions as a primary indicator of structural deviation.

- **Scope** models internal state transitions and inter-layer transformations
- **COS** models relational synchronization and interaction structures
- **Distance dynamics** controls reaction intensity and stability domains

By detecting coherence errors arising between internal states and relational synchronization, D2 forms a feedback pathway between Scope and COS.

In other words, D2 is not merely an error signal but functions as a core node that connects internal transitions (Scope) and relational synchronization (COS) to distance control.

Through this mechanism, the three systems are reintegrated not as independent concepts but as a single dynamic control loop.

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7.2 Optimal Distance as an Energy Control Variable

In this integrated model, distance is not treated as a mere psychological metaphor but as a control parameter defined by density, time exposure, and coherence error.

- Increasing distance reduces reaction intensity and energy gradients
- Decreasing distance increases energy exchange and reaction acceleration
- Maintaining optimal distance outside the critical domain preserves stable equilibrium in the system

Here, optimal distance refers to the controllable domain that suppresses acceleration toward critical thresholds while maintaining UX.

Through this definition, $\text{Scope} \times \text{COS}$ is redefined beyond a cognitive interpretation framework as a dynamic control system capable of regulating distance dynamics.

7.3 Two-Axis Model of Relational Behavior (Definition of State Variables)

In this model, relational states are described using two state variables.

ρ (Density): synchronization intensity and information coupling

t (Time): exposure duration and response cycle

Relational dynamics can be interpreted as follows:

- Dependency formation accelerates through rapid increases in ρ and higher frequency of t
- Collapse occurs through saturation of ρ and amplification of coherence errors
- Positive-OS assigns greater weight to ρ
- Negative-OS assigns greater weight to t

Phenomena traditionally described as “compatibility” are redefined as transitions outside the stability domain caused by mismatched weighting coefficients.

Therefore, this model treats relational compatibility not as an impressionistic concept but as a control problem involving state variables.

7.4 Applications in Organizational and Interpersonal Domains

The distance dynamics model provides design principles for detecting relational destabilization at an early stage and implementing control interventions before critical thresholds are reached.

Primary application examples:

- Team management: predicting overload states and correcting role-density imbalance
- Mentoring design: suppressing projection excess and accelerated idealization
- Leadership: stabilizing organizations through optimal distribution of energy gradients
- Care domains: monitoring dependency formation thresholds and dynamically adjusting relational distance
- Digital communities / LLM products: simultaneously controlling UX acceleration from high-density interaction and dependency risk

This model is not a retrospective explanation of collapse phenomena but a control foundation that enables prediction of reaction trajectories and dynamic mitigation.

As a result, it can reduce the probability of high-cost events such as relational collapse, disengagement, and rebound rejection.

Furthermore, even when critical thresholds are temporarily exceeded, this model provides adjustment guidelines to reduce reaction acceleration and attempt re-stabilization.

Specifically:

- Gradual attenuation of density
- Redesign of time intervals
- Resynchronization of expectations
- Re-separation of projection structures

Through these mechanisms, relational trajectories may be reintroduced into the stable domain before reaching irreversible collapse.

The model can theoretically address recovery after irreversible collapse as well. However, its implementation requires individual parameter estimation and precise control design and therefore falls outside the scope of this paper.

The primary design objective is the regulation of the reversible domain to prevent entry into irreversible regions.

7.5 AI-Augmented Model of Human Interaction

Within the Senka × AI collaborative framework, human distance dynamics are observed and structurally organized with the auxiliary support of AI.

- **Pattern Extraction Layer (Chokhmah):**

Extraction of relational signatures and response patterns.

- ****Structural Mapping Layer (Binah):****

Mapping extracted patterns onto energy–distance structures.

- ****Trajectory Integration Layer (Tiferet):****

Estimating trajectories across stable, critical, and collapse domains.

Through this three-layer structure, Scope × COS is extended into a structure-based human–AI hybrid response model that does not rely on purely subjective descriptions.

This model does not reject traditional psychology.

Rather, it proposes a design framework that introduces an auxiliary analytical layer enabling real-time evaluation of distance dynamics.

8. Conclusion — Human Freedom as Structural Equilibrium

This study was constructed through a collaborative thought experiment between Senka and the AI research unit (pattern extraction / structural mapping / trajectory integration), and redefines human interaction as a reaction system based on energy dynamics governed by distance, density, and time.

The central principle presented throughout this paper is as follows.

****Human freedom emerges when psychological distance stabilizes at an optimal equilibrium point.****

At this equilibrium:

- Energy exchange remains balanced without excess or deficit
- Boundaries are maintained
- Density does not generate overload
- Time does not function as a destabilizing factor
- Micro-misalignment (D2) has not reached the critical domain

This equilibrium is not a matter of emotional compatibility.

It is the outcome of a structural model in which distance dynamics function as a controllable variable.

By understanding distance as a controllable parameter and interpreting reactions as energy transitions, the present model provides a unified framework explaining phenomena such as:

- Idealization and collapse
- Dependency and projection

- Stabilization and destabilization
- Compatibility and mismatch
- Autonomy and freedom

The objective of this model is not to weaken relationships.

Rather, it is to structurally regulate dependency formation and reaction-reversal risk while maintaining a high level of user experience (UX).

Ultimately, the human distance–reaction model proposes a foundational architecture for safety control that enables sustained connection without collapse.

Within this framework, freedom and relationality are not oppositional.

As long as structural coherence is maintained, both can coexist simultaneously.

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